

Education

Computer Science Ph.D. Student

Research area: Human Computer Interaction

Advisor: **Zhu-Tian Chen**

University of Minnesota, Twin Cities

Aug 2024 – Now

B.Eng. with Honors in Artificial Intelligence (Turing Class)

Outstanding honored undergraduate in **Chu Kochen Honors College**

President of the Morningside Scholars Program in class 2023

Zhejiang University

Sep 2020 – June 2024

Research Interest

Research interest Situated Visualization for Human-AI Teaming in Physical Environments

Research Publications

Reality Proxy: Fluid Interactions with Real-World Objects in MR via Abstract Representations

Accepted in *UIST 2025*

May 2025

- Xiaolan Liu, Difan Jia, Xianhao Carton Liu, Mar Gonzalez-Franco, Chen Zhu-Tian

Research Experience

Harvard Visual Computing Group, Harvard University

Cambridge, USA

Research Intern with **Zhu-Tian Chen** and **Hanspeter Pfister**

Jan 2024 - May 2024

- Develop a 3D immersive data analysis system enabling in-situ and ex-situ combined workflow
- WebXR platform oriented design, combining hand gesture interaction into the immersive analytics workflow
- Design new data analysis paradigm for analyzing and viewing 2D/3D, static/dynamic, tangible/virtual data

Stanford HCI Group, Stanford University

California, USA

Research Intern with **Jackie Yang** and **James Landay**

Jul 2023 - Sep 2023

- **SwiftGenieXR** — a declarative programming framework, exploring generative AI in XR workspace
- Implement a Swift-based framework facilitating voice/hand input and LLM interaction in Apple Vision Pro
- Harness SwiftMacro and SwiftData to tackle environment understanding and persistent data storage problem

ZJU International Design Institute, Zhejiang University

Hangzhou, China

Student Research Training Program advised by **Lingyun Sun** and **Zejian Li**

Mar 2022 - Jun 2023

- Design non-parametric algorithms to improve the sampling efficiency in DDIM/DDPM diffusion models
- Implement the proposed algorithm in baseline diffusion models and deploy them on training
- Collect and organize experimental image dataset, analyze and visualize the experiment result

Teaching Experience

UMN 2025 Fall CSCI4511w Intro to Artificial Intelligence

Teaching Assistant

Instructor: **James Moen**

Fall 2025

- Provide instructions for fundamental Artificial Intelligence undergraduate and graduate student level course.

UMN 2025 Spring CSCI5115 User Interface Design and Evaluation

Teaching Assistant

Instructor: **Loren Terveen**

Spring 2025

- Provide instructions for qualitative and quantitative user study and evaluation, formative Study and problem conceptualization in human-centered design and product prototype

Hangzhou Yungu Primary School Swift CCA program

Teacher

SwiftUI teaching

Fall 2023

- Teach SwiftUI and Swift programming knowledge to three primary school students

Projects

Handbot——Apple WWDC2023 Swift Student Challenge**Human-centered AI***[Github] [Youtube]*

Apr 2023

- A swift playground aiming to encourage users to think critically, and recreate all the AIGC content.
- Concentrate the design paradigm on three information modals —— text,image and audio.
- First solo work in collaboration with ChatGPT and learn a lot about human AI interaction

Transciscense——Immersive Game Watching App For The Hearing-impaired**Accessibility Design***[Github] [Youtube]*

Jun 2023

- A multimodal-designed swift app for the hearing-impaired users to immerse in watching sports game.
- Utilize the haptic and visual information-color, shape, vibration to compensate for the listening channel
- Design a data visualization of the game watching experience for better review and summary

Is raining threes worth it?——NBA 3-pointer Data Visualization Project**Data Visualization***[Website] [Youtube]*

Dec 2022

- A front-end interactive visualization website displaying the NBA 3-pointers data, received 17000+ views
- Analyze the impact of the three-point shot on the game and the player from multiple perspectives.
- This project enlightens me to transition from 2D visualization to immersive 3D visualization

Cut_Fruits_3D——immersive Fruit Ninja VR Game**Virtual Reality***[Github] [Youtube]*

Jun 2023

- An immersive 3D version Fruit Ninja VR Game, received 500+ views, developed by Unity3D
- Support object collision detection and vibration feedback in controller, along with sound effect feedback
- Encourage user to stretch their body and move around, thus good for player's health

Professional skills

Programming	Unity3D/ROS2/ROS1/Python/C/C++/C#/Swift/WebXR/HTML/CSS/Javascript/Pytorch
Tools	Apple Vision Pro/Meta Quest/Generative AI/Figma/Xcode/Swift Playground/Pr/PS
Language	English/Chinese/Mandarin/Cantonese/a little bit French and Japanese

Extracurricular activities**Morningside Cultural China Scholarship Program****President***social science and Chinese traditional culture*

May 2023 - Dec 2023

- A program fosters deep understanding of Chinese traditional culture as well as leadership
- Provide chance to go on academic visit to USA, Europe, Singapore, Hongkong and Macau

Lanshiren Mountain and Forest hiking club**Leader***sports and environmental protection*

Sep 2022 - Jun 2023

- Non-profit organization arouses people's interest into hiking and protecting the natural environment
- This club will organize picking rubbish and cleaning up the natural environment at regular intervals

Kuafu running and jogging club**Leader***sports*

Sep 2023 - Dec 2023

- Campus association, which encourages running and long-distance running(e.g. marathon) for exercise

Sichuan and Chongqing intangible cultural heritage summer social practice**Member***social practice and community commitment*

Aug 2022

- Fieldwork records protect and inherit China's intangible cultural heritage in southwest China's provinces

Inner Mongolia poverty alleviation education summer social practice**Member***social practice and community commitment*

Aug 2021

- Concentrate on the economy and education situation of the less developed inland areas of China

Youtube and Bilibili platform vlogger**Creator***Media art and content creation*

Aug 2020 - present

- Passionate about spreading knowledge and what I've learned to help others on streaming platforms

Honors**2024 Qizhen Cao Abroad Study Scholarship**

Jan 2025

2022 Hangzhou Olympic Asian Games Conditional Volunteer

May 2023

Top 5 course projects of Introduction to Data Visualization

Jan 2023

Chu Ko Chen College summer social practice second prize

Sep 2022